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Contrasting epigenetics of *Ixodes scapularis* populations

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Hard ticks are a source of public health concern, in part due to their ability to inhabit different environments. In the United States (US), blacklegged ticks (*Ixodes scapularis* Say), the primary vector of Lyme disease, exhibit various phenotypes depending on their geographic origin (i.e. northern and southern US ticks). Although genetics may partially explain how blacklegged tick populations acclimate to different environmental conditions across the US, epigenetics may also contribute to their success. Epigenetic mechanisms, such as DNA methylation, might modulate gene expression allowing for rapid adaptation. To gain insight into the potential contribution of DNA methylation, an Enzyme-Linked Immunosorbent Assay (ELISA) was utilized to evaluate differences in DNA methylation levels between blacklegged ticks collected from Minnesota (northern region) and Texas (southern region). DNA methylation profiles from both populations were characterized using bisulfite and nanopore sequencing. Our results revealed significant variability in global methylation levels between southern and northern tick populations, as well as highly variable relative expression of genes encoding DNA methyltransferases and demethylases. Overall, northern blacklegged ticks exhibit lower global DNA methylation levels than southern ticks. Basic proline-rich protein, zinc finger protein 501-like protein, and an uncharacterized protein LOC115333191 are among the genes that exhibit lower DNA methylation. Our findings revealed that blacklegged tick populations possess distinctive DNA methylation profiles, which may contribute to their phenotypic plasticity across the US. This study aims to pave the way for future research into the potential molecular mechanisms that allow ticks to successfully acclimatize to environmental changes.

Keywords DNA methylation, epigenetics, ticks, DNA methyltransferases

Lyme disease (LD) is the most prevalent vector-borne disease in the United States (US), with 500,000 estimated cases each year[1–3] and an estimated economic burden of approximately $968 million annually[4]. In the eastern US, LD is primarily transmitted by the blacklegged tick, *Ixodes scapularis* Say. This tick is a three-host tick, capable of transmitting and harboring a wide array of pathogenic microorganisms, including *Borrelia burgdorferi* (the main causative agent of Lyme disease), *Anaplasma phagocytophilum*, and *Babesia microti*[5–7]. Blacklegged ticks can be co-infected with various pathogens simultaneously, which represents a risk for co-infections through the transmission of multiple pathogens in a single bite[8,9]. The lack of efficient tick control methods[3,10], ongoing geographic expansion of tick populations[11–13], and their ability to carry multiple zoonotic pathogens[5–7,9,14,15] collectively contribute to the growing health concern posed by this tick species.

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Blacklegged ticks have been reported throughout much of the eastern and upper midwestern US[13]. Interestingly, the incidence of pathogens associated with blacklegged ticks shows a disproportionate geographic distribution, with higher transmission rates in northern than in southern states[13,16–18]. The molecular mechanisms underlying these dissimilarities remain unknown, although genetic variation is suspected. Two primary haplotypes have been identified in northern and southern blacklegged tick clades, within these populations, additional haplotypes have arisen depending on the genes that are analyzed. The northeastern and upper midwest haplotypes are more genetically similar and are associated with increased tick-borne disease occurrence, whereas the southern group exhibits greater haplotype diversity and a lower pathogen infection rate[17,19,20]. Further, within the northern and southern populations, subpopulations appear to differ in their ability to acquire infection. For example, Xu et al., 2020[17] showed that only *I. scapularis* belonging to mitochondrial haplotype A were infected with *B. burgdorferi*, whereas haplogroup S individuals were uninfected. Whether these haplotypes reflect heterogeneity in vectorial capacity remains to be determined experimentally. Previously, variation in pathogen transmission rates between the northern and southern US were attributed to divergences in vector competence between two presumably different *Ixodes* species (*Ixodes dammini* [found in the northern US] and *I. scapularis* [found in the southern US]). Yet, vectorial capacity experiments demonstrated that southern clade ticks are just as effective in acquiring and transmitting *B. burgdorferi* from and to mice as northern clade ticks, despite northern ticks being more efficient at feeding on mice[21]. Subsequent mating experiments and phylogenetic analyses determined that these were conspecific populations, resulting in their synonymization to *I. scapularis*[21–23].

Southern and northern ticks differ in seasonal activity[24], infection rate[25], host association[26,27], and host-seeking behavior[28,29]. Nonetheless, geographic location and the vertebrate host species are the strongest predictors of LD incidence[27]. Specifically, northern ticks are typically found feeding on *B. burgdorferi* competent mammalian hosts, whereas southern ticks prefer incompetent reptilian hosts[26]. These host associations may drive natural variation in questing behavior, with southern immature life stages maintaining a lower questing height, often remaining beneath the leaf litter. Conversely, northern immatures are more commonly found above the leaf litter, actively searching for hosts[28,30]. These variations in host-seeking behavior and host preference may partially explain the differences in nymphal infection prevalence and tick-borne pathogen transmission between the southeastern and northern US regions[26,28,29].

However, the genetic and molecular mechanisms underlying the observed phenotypic divergence among blacklegged tick populations remains unknown. For instance, epigenetic mechanisms, such as DNA methylation, has been documented in the genome of blacklegged tick embryonic cell lines (IDE2, IDE8, and ISE18)[31]. In arthropods, DNA methylation is associated with phenotypic plasticity in response to environmental cues such as temperature[32–34]. Further, recent assembly and annotation of the *I. scapularis* genome revealed the presence of epigenetic clusters (polycomb and trithorax group proteins) involved in transcriptional regulation[35]. Whether DNA methylation plays a role in blacklegged tick adaptation to environmental conditions (e.g., temperature) remains understudied[33,34,36]. Given the potential link between DNA methylation and tick phenotypic plasticity, we investigated (i) global DNA methylation levels, (ii) differentially methylated regions (DMRs), and (iii) the relative expression of DNA methylating and demethylating enzymes in two blacklegged tick populations, one representing northern populations [Minnesota] and another for southern populations [Texas] of the US. Given that these populations experience different ecological conditions, we hypothesized that blacklegged ticks from the north and south differ in the level of 5-methylcytosine (5-mC) methylation within their genomes, and that this variation is concentrated at specific loci. Our aim was to determine whether differences in DNA methylation correspond to geographic origin, particularly among southern and northern blacklegged tick populations. Our results confirmed that ticks collected from Minnesota (MN) and Texas (TX) contain distinct 5-mC profiles that vary according to sex, and geographic location. These DNA methylation profiles provide evidence of epigenetic plasticity among natural *Ixodes scapularis* populations, which may contribute to their ability to cope with and respond to fluctuations in environmental conditions.

Results

The percentage of 5-methylcytosine (%5-mC) varies among *Ixodes scapularis* populations across different years and geographical regions

Previous studies have reported genetic variation between northern and southern blacklegged ticks[17,37]. To assess whether genetic variation between these populations also includes differences in DNA methylation, we quantified the level of 5-mC in MN and TX blacklegged tick populations using ELISA assays. Our preliminary results showed significant variation in methylation levels between ticks from MN and TX, with MN ticks displaying higher methylation levels than TX ticks ($p = 0.0015$). Nevertheless, due to the few ticks tested, these experiments were repeated to confirm the results.

We performed further testing using ticks collected in 2021 and 2022 to determine whether these variations were replicable. We observed high variability in %5-mC between groups during two distinct years (Supplementary Fig. S1). Notably, no significant differences were found between MN and TX ticks collected in Spring 2021 ($p = 0.849$) (Fig. 1a, Supplementary Fig. S1). However, significant variation in the %5-mC was observed between MN and TX ticks collected in Spring 2022, using unpaired two-sided t-test (2022; $t\ (30) = 4.568$, $p < 0.0001$) (Fig. 1b). Moreover, within each location, methylation levels varied between tick sexes (Supplementary Fig. S1). Males from MN displayed a higher %5-mC in their genome when compared to males from TX, regardless of season or collection year (Fig. 1c, Supplementary Fig. S1e, f). Thus, our results suggest that global DNA methylation in cytosines is highly variable in female *I. scapularis*, but more conserved in males.

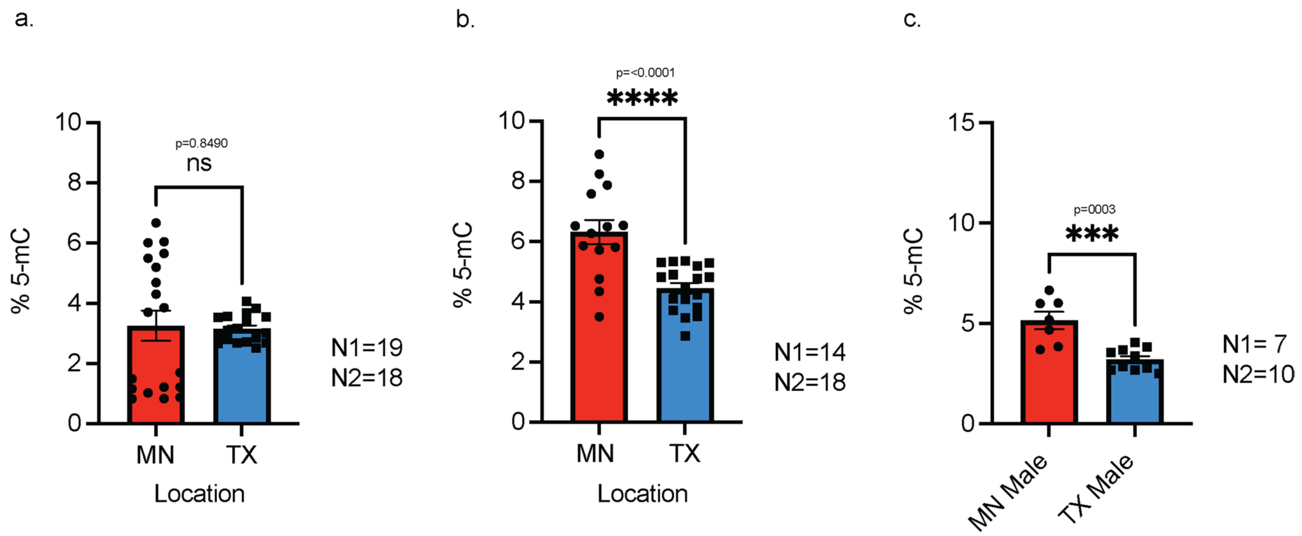

Fig. 1. Percentage variations of 5-mC between blacklegged ticks collected over a span of 2 years. Comparison of the methylation percentage variations in cytosine residues among MN and TX ticks collected in (**a**) 2021 and (**b**) 2022 [both sexes combined]. (**c**) Represents the dissimilarities in cytosine methylation among MN versus TX tick males collected in 2021. Significant differences were evaluated with an unpaired two-sided t-test. The error bars represent the mean ± SEM of the two variables. Asterisks indicated significant variability between samples. Abbreviations = MN: Minnesota, TX: Texas, and percentage of 5-mC: %5-mC. Representative figures from four experiments are shown. N1 = number of MN ticks and N2 = number of TX ticks used for comparison. The results from all experimental replicates are provided in Supplementary Fig. S1.

Protein	Organism	Best Hit	I. scapularis accession number	Query cover (%)	E-value	Percentage of Identity (%)	Size protein (aa)
DNMT1	*Homo sapiens*	DNA (cytosine-5)-methyltransferase PliMCI	XP_029831274	76	0.0	57.50	1,363
DNMT2	*Homo sapiens*	tRNA (cytosine(38)-C(5))-methyltransferase [*I. scapularis*]	XP_029848442	99	6e-91	42.09	361
DNMT3A	*Homo sapiens*	DNA (cytosine-5)-methyltransferase 3 A isoform X1 [*I. scapularis*]	XP_040064066	44	7e-96	48.03	363
TDG	*Mus musculus*	G/T mismatch-specific thymine DNA glycosylase [*I. scapularis*]	XP_029824608	57	1e-78	55.97	705
TET3	*Homo sapiens*	uncharacterized protein LOC8042599 isoform X1	XP_029838410	34	1e-118	48.34	2,116
		uncharacterized protein LOC8042599 isoform X2	XP_040075936	36	9ee-119	48.34	2,090
		uncharacterized protein LOC8042599 isoform X3	XP_040075937	34	1e-118	48.34	2,062
		uncharacterized protein LOC8042599 isoform X4	XP_040075938	34	8e-119	48.34	2,036
		uncharacterized protein LOC8042599 isoform X5	XP_029838415	32	2e-119	48.34	1,767
		uncharacterized protein LOC8042599 isoform X6	XP_040360312	32	2-e-119	48.34	1,713

Table 1. Blacklegged tick DNA methyltransferases and demethylases identified by PSI-BLAST.

Identification of genes encoding enzymes putatively involved in DNA methylation homeostasis in the blacklegged tick

Currently, the genes encoding DNA methyltransferases (DNMTs) and thymine DNA glycosylase (TDG) within the genome of the blacklegged tick have been annotated using gene prediction[38]. To confirm the identity of DNMTs and demethylases of this species, we conducted a homology search using PSI-BLAST with human and mouse homologs. These species were chosen as they possess a complete set of DNMTs, unlike other arthropods, such as *Drosophila*, which possess only one DNA methylation enzyme (homolog of DNMT2)[31]. We identified a complete set of homologs for three DNMTs, along with the ten-eleven translocation (TET) 3 and TDG enzymes, known to catalyze demethylation. The top homologous proteins found in the *I. scapularis* genome exhibited identity percentages ranging from 57.5 to 42.0%, when compared to the human homologs, and query coverage between 32 and 99% (Table 1).

To validate that the identified genes encode homologs to these enzymes, the presence of conserved domains within each putative blacklegged tick homolog was confirmed using the normal mode in SMART[39]. A potential homologous DNMT1 from the blacklegged tick, abbreviated as IscDNMT1, encoded a full-length protein of 1,363 amino acids, displaying four distinct domains: (1) the replication foci domain (DNMT1-RFD) facilitating methylation at the correct residue (positions 160–293); (2) a zinc-finger domain (zf-CXXC) that binds to an unmethylated CpG site (positions 396–442); (3) two consecutive bromo adjacent homology domains (BAH) typical of DNMT1 (positions 513–642 and 694–863) associated with DNA (cytosine-5) methyltransferases; and

(4) the DNA methylase domain (c-5 cytosine-specific DNA methylase) responsible for methylating the fifth carbon of cytosine in DNA and producing the modification C5-methylcytosine (positions 902-1,355; Fig. 2a). On the other hand, IscDNMT2 encoded a 361 amino acid polypeptide, and IscDNMT3 exhibited a full-length chain of 363 amino acids, both displaying a unique DNA methylase domain in their polypeptide sequences at positions 19–357 and 58–235, respectively (Fig. 2b, c).

In the case of IscTET3, this demethylase consisted of six isoforms ranging in length from 1,713 to 2,116 amino acids (Table 1). Each isoform features an oxygenase domain (Tet_JBP) responsible for catalyzing the conversion of 5-mC into 5-hydroxymethylcytosine (5hmC), followed by subsequent 5-formylcytosine (5fC) and 5-carboxylcytosine (5caC). The Tet_JBP domain is consistently positioned at the end of the sequences alongside a DNA binding domain (PBD/zf-CXXC), which is absent in isoforms X5 and X6 (Fig. 2d, Supplementary Fig S2). IscTDG encoded a protein of 705 amino acids with three primary domains: (1) SCOP d1lsha3 domain, serving an unknown function, located at position 71–111, (2) three DNA binding domains (AT hooks) positioned at 158–170, 188–200, and 509–521, and (3) an Uracil-DNA glycosylase domain (UDG), a member of DNA repair enzymes responsible for actively removing products generated by TET at position 247–440 (Fig. 2e). Therefore, *I. scapularis* possesses a complete set of enzymes maintaining DNA methylation homeostasis.

Female ticks from TX had higher DNMT1 expression than those from MN, while males showed no significant difference

To investigate whether the observed variation in 5-mC levels correlate with methylase and de-methylase activity, we used qRT-PCR to estimate the relative expression of blacklegged tick DNA methyltransferases and demethylases between populations in MN and TX. The expression levels of DNMT2, DNMT3, TET3, and TDG showed high variability among biological replicates (inter-replicate *CV* range: 27%–154%). We detected significant differences between populations and sexes, although these patterns were inconsistent across replicates (Supplementary Fig. S3). Notably, females from TX showed a significantly increased expression of DNMT1 when compared to females from MN (Fig. 3; $p = 0.0008$). These females also presented markedly higher levels of DNMT1 expression when compared to males from both regions (Fig. 3; TX $p = 0.0184$, and MN $p < 0.0001$). In contrast, no significant differences in expression were found between males from MN and TX ($p = 0.236$). These results indicate that, under these conditions, only DNMT1 presents a significant discernable expression pattern.

Analytical approaches uncover variable but opposite 5-mC trends in MN and TX ticks

To analyze 5-mC in the CpG context, the most prevalent form of DNA methylation in arthropods[40–42], we employed two complementary yet methodologically distinct approaches: whole genome bisulfite sequencing (WGBS), a short-read technology applied to pooled samples, and nanopore sequencing, a long-read technology applied to individual samples. DMRs were subsequently identified within genic and predicted promoter regions.

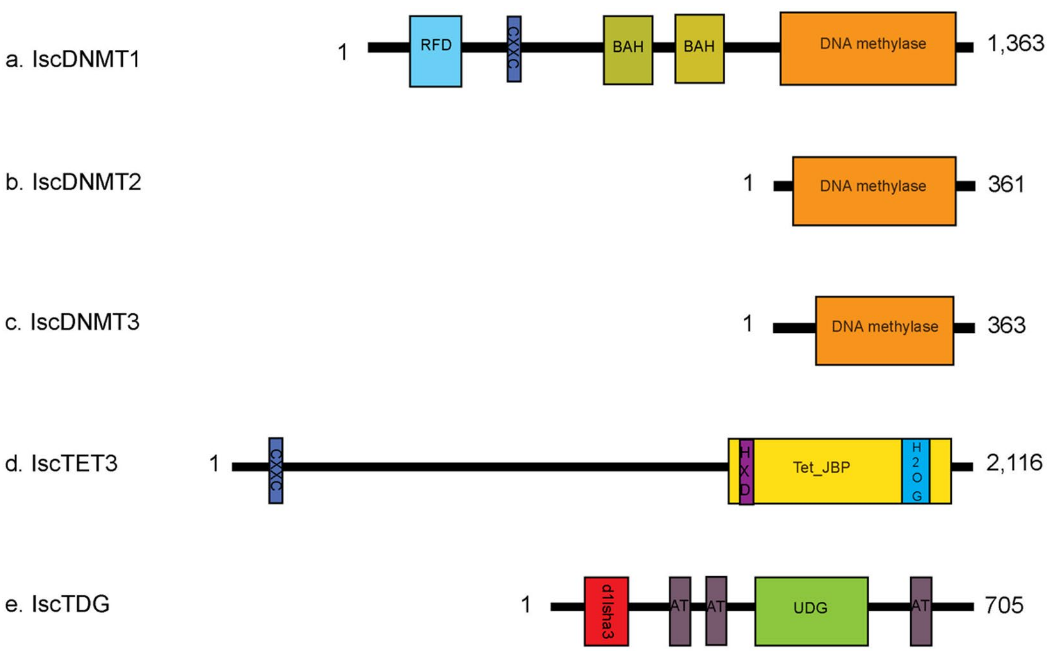

Fig. 2. Domains identified in *Ixodes scapularis* DNA methyltransferases and demethylases homologous. Conserved domains and binding residues typical in (**a**) DNMT1, (**b**) DNMT2, (**c**) DNMT3, (**d**) TET3 and (**e**) TDG were detected in blacklegged tick homologs identified by PSI-BLAST. RFD = replication foci domain; CXXC = zinc-finger domain; BAH = bromo adjacent homology domain; HXD = HXD Fe^{2+} binding residues; H2OG = H Fe^{2+} and 2-Oxoglutarate binding residues; Tet_JBP = oxygenase domain; d1Isha3 = SCOP d1lsha3 domain; AT = AT hooks; UDG = uracil-DNA glycosylase domain. Numbers represent the protein size in amino acid residues (for TET3, we are displaying the amino acid size for isoform 1.

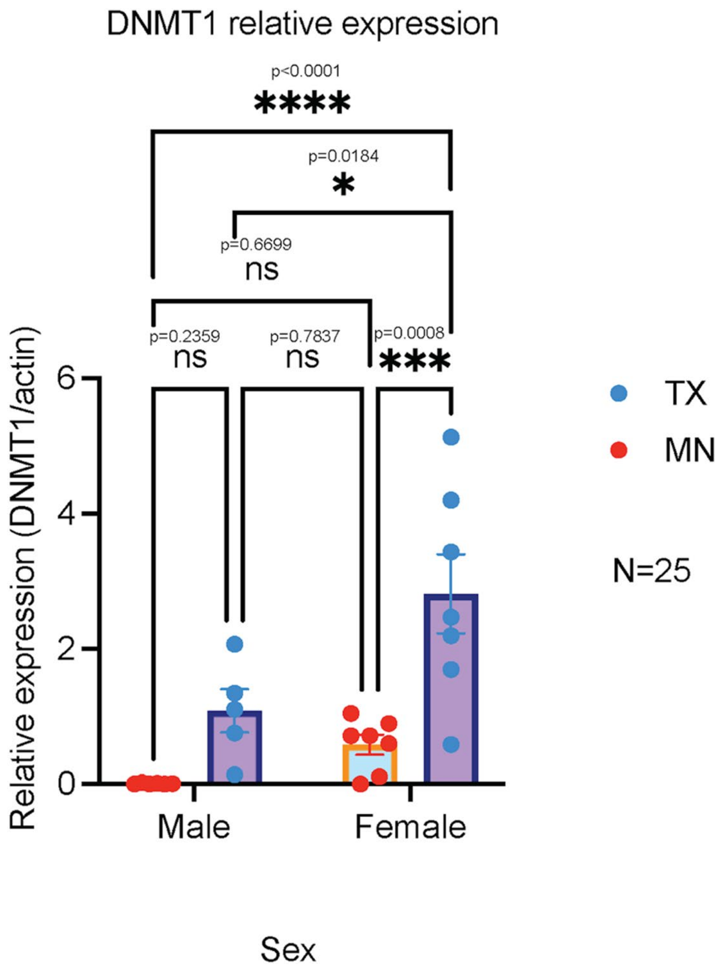

Fig. 3. DNMT1 is upregulated in female ticks collected in Texas. TX females displayed statistically higher relative expression of DNMT1 compared to males from TX, males from MN, or females from MN. Differences in expression were evaluated using a Two-way ANOVA followed by Tukey's test for multiple comparisons. Asterisks indicated significant differences between samples. Abbreviations: MN: Minnesota and TX: Texas. Number of ticks compared: MN female = 7, MN male = 6, TX female = 7, and TX male = 5. A representative figure from four experiments is shown. The results from all experimental replicates are provided in Supplementary Fig. S3.

To ensure analytical reliability, sequencing quality was assessed and found to be high across both platforms. The average bisulfite conversion rate for WGBS was 99.69%, and CpG coverage averaged 17.94X and 16.34X for MN and TX, respectively. Nanopore sequencing achieved comparable CpG coverage of 11.53X and 10.83X for MN and TX, respectively. These coverage levels were supported by high sequencing depth, with a mean number of raw sequences obtained from pooled females and males of 220,661,496 for MN and 332,932,785 for TX.

After quality control, the mean number of sequences was 215,612,600 (27.39 Gbp) for MN and 324,838,053 (41.26 Gbp) for TX (Supplementary Table S1). These reads were then mapped to the reference genome (NCBI, ASM1692078v2), yielding an average mapping efficiency of 25.2% for MN and 31.7% for TX (Supplementary Table S2, S3). For nanopore sequencing, the mean number of raw sequences obtained was 17,259,423 (29.58 GB) and 9,431,732 (25.88 GB) for individual adult females from MN and TX, respectively. Mapped sequences achieved efficiencies of 97.93% and 98.20% for MN and TX, respectively (Supplementary Table S4).

To characterize the DNA methylation in the MN and TX populations, we examined DNA methylation levels in distinct contexts using WGBS, a method that has previously been used in ticks[40]. Genome-wide bisulfite sequencing revealed that the 5-mC in the CpG context was the most abundant, with an average of 9.74% for MN ticks and 12.9% for TX ticks (Fig. 4a). Additionally, TX females exhibited statistically higher levels of 5-mC in the CpG context compared to TX males, MN females, and MN males, while TX males had higher CpG 5-mC compared to MN males and females (Fig. 4b, $p < 0.0248$). Notably, TX ticks exhibited increased levels of 5-mC compared to MN ticks (Fig. 4a; t (9) = 6.322, $p = 0.0001$). The next most common context was CHH, showing 1.24% and 1.25%. This was followed by CHG with 1.14% and 1.15% in MN and TX. However, no meaningful variation in DNA methylation was observed among populations in any of these contexts (Supplementary Table S2, Supplementary Fig. S4a, b; $p = 0.733$).

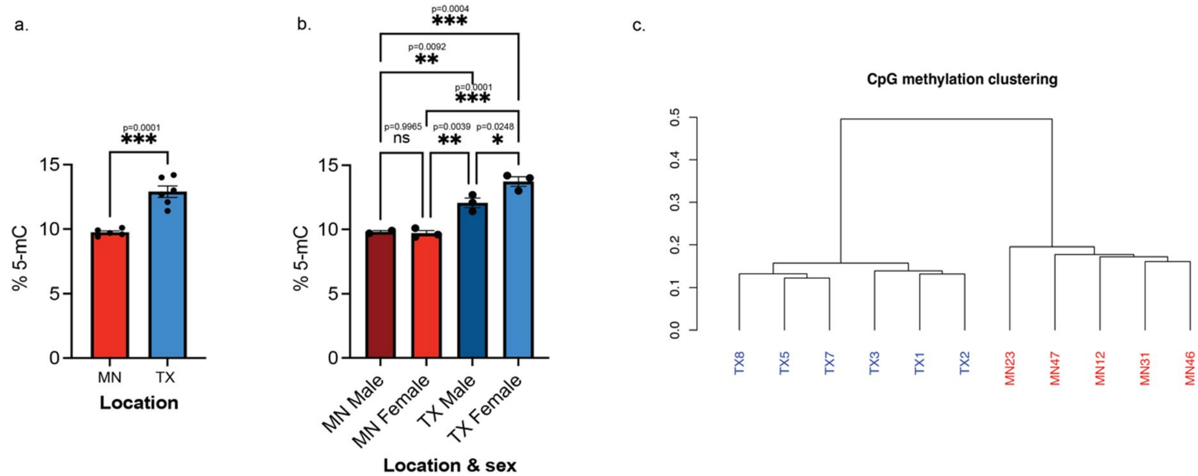

Fig. 4. MN versus TX ticks present variable methylated level. Differential DNA methylation level in CpG context between ticks from both states, identified by whole bisulfite sequencing (WGBS), were tested using unpaired two-sided t-test (**a**) and ordinary one-way ANOVA (**b**), both of which revealed statistically significant differences. Additionally, a comparative analysis was conducted using DNA methylation residues of blacklegged tick populations. Cluster analysis based on correlation distance indicated a distinct separation of the DNA methylation profiles of ticks collected in MN or TX (**c**). Significant differences were displayed in asterisks. Error bars represent the mean ± SEM of the two variables. The "Y" axis always represents the percentage of DNA 5-methylcytosine (%5-mC). Abbreviations: MN: Minnesota and TX: Texas.

Additionally, pooled females and males from WGBS were used for comparative analysis to identify differentially methylated genes in CpG sites among both blacklegged tick populations. Using a correlation distance method that coincides with principal component analysis, we detected two distinct clusters that represents unique variation in methylation profiles between MN and TX blacklegged tick populations. Notably, the CpG methylation profiles from TX ticks were more distinct from one another and showed clear sex-based clustering (Fig. 4c, Supplementary Fig. S4c). Differentially methylated CpG sites were detected among MN and TX populations, with the most abundant genes being hypomethylated bases (Fig. 5). For instance, blacklegged ticks from TX showed greater variation in methylated CpG sites within genic regions compared to MN ticks. A total of 1,073 and 14,675 hyper- and hypomethylated sites (associated with 109 and 704 genes), respectively, were identified in the analysis comparing MN to TX ticks. Many of these sites are distributed across the largest scaffolds in the blacklegged tick karyotype, such as NW_024609839.1 and NW_024609857.1 (Fig. 6).

Variation in DNA methylation between blacklegged tick in the north and south may result from epigenetic and genetic differentiation[43,44]. This latter mechanism can be traced by examining single nucleotide variation[43]; therefore, we explored variable sites in males and females from both locations using whole genome sequencing (WGS). Females and males from both locations were assigned to one population for this analysis. The mean number of raw sequences was 333,465,834 bases (33.47 Gbp) and decreased to 333,465,831 (32.52 Gbp) after quality control. Filtered sequences were then mapped to the reference genome (NCBI, ASM1692078v2) with a mean percentage of mapped reads of 97.51% (Supplementary Table S5). In total, 143,654,820 variable sites were detected from MN and TX ticks together, and after filtering, the total number of single nucleotide polymorphisms (SNPs) were 19,799,421 (Supplementary Table S6). This process led to the removal of 14 differentially methylated bases (DMBs) and nine genes from the final bisulfite hypermethylated list due to the presence of SNPs (Supplementary Table S7). For its counterpart, the hypomethylated list, 57 DMBs were identified, and 10 genes were excluded because their DMBs overlapped with SNPs (Supplementary Table S8). Additionally, twenty genes were identified that contained both hyper- and hypomethylated marks from WGBS (Supplementary Table S9). The final lists included 78 hypermethylated and 668 hypomethylated genes from WGBS (Supplementary Table S10, S11).

On the other hand, nanopore sequencing revealed an opposite trend, where female ticks from MN showed greater methylation in genic regions than female ticks from TX. In total, nanopore-based analysis identified 114 genic DMRs between these two populations of blacklegged ticks, with 63 of these DMRs corresponding to unique protein-coding genes (Supplementary Fig. S5). Six loci overlapped with either hyper- or hypomethylated genes identified via bisulfite sequencing (Supplementary Table S12). After removing the overlapping genic regions, 30 hyper- and 28 hypomethylated genetic regions were obtained from the nanopore sequencing data (Supplementary Table S13, S14).

Overall, despite revealing an opposing genome-wide methylation trend between sequencing modalities, the long-read sequencing corroborated hypermethylation at three specific gene transcripts in TX ticks that were also identified by WGBS (XM_029973219.3, XM_040221513.2, and XM_040221333.3) (Supplementary Table S11, S13).

Promoter methylation species was also analyzed from the differentially methylated sites. A total of 13 hypermethylated and 65 hypomethylated promoter regions were found in MN ticks compared to TX ticks. These

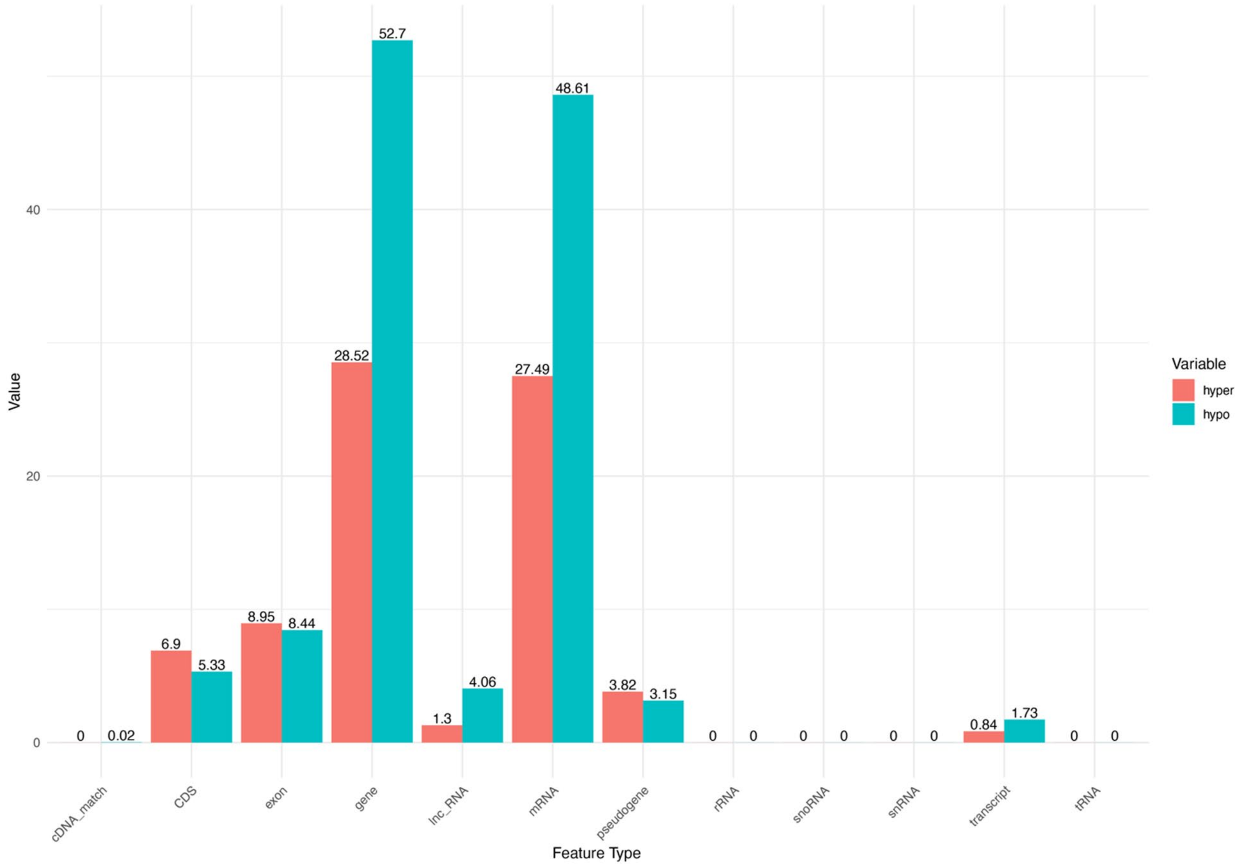

Fig. 5. Hypomethylated bases are predominantly present within genic regions. Hyper- and hypomethylated bases were plotted by genomic features. Salmon bars indicate hyper- DNA methylation in MN ticks relative to TX ticks (levels of methylation in MN/levels of methylation in TX), whereas hypomethylation is represented with dark turquoise bars. The numbers over the bars represent the percentage of differentially methylated bases within each genomic feature. Abbreviation hyper: hypermethylated sites and hypo: hypomethylated sites.

hypermethylated sites were in the predicted promoter regions of genes encoding microtubule-associated protein 4 (XM_040210232.3), semaphorin-2 A (XM_029994482.4), and zinc finger proteins (e.g., XM_040210244.2). Several potential transcription factors (TFs; XM_042294713.1, XM_040217705.3, XM_040217697.1, XM_029994269.4) in ticks were also found to have hypomethylated residues within their promoters, as well as other genes that encode TIR transposons (XM_040205953.1, XM_040211267.1 and XM_040209593.3), a histone (XM_029973769.4), and serine/arginine-rich splicing factor 2 (XM_029967191.4), among others (Supplementary Table S15). Overall, DMRs varied between TX and MN populations, although the tendencies were inconsistent among approaches.

Overrepresented gene ontology (GO) terms in hypomethylated genes

To assess the potential biological significance of DMRs identified between northern and southern populations of blacklegged ticks, we performed enrichment analysis based on GO, proteins, and pathways. Hypomethylated genes were overrepresented in 203 GO terms. Enriched biological processes included the regulation of transcription by RNA polymerase II (GO:0006357), regulation of gene expression (GO:0010468), regulation of DNA-templated transcription (GO:0006355), and macromolecule biosynthetic process (GO:0009059). Additionally, response to oxidative stress (GO:0006979) was identified as another major biological process category (Supplementary Table S16, Supplementary Fig. S6). The most prominent cellular components were the endomembrane system (GO:0012505), cell periphery (GO:0071944), and cytoskeleton (GO:0005856) (Supplementary Table S17, Supplementary Fig. S7). Among molecular functions, transcription regulator activity (GO:0140110) and protein binding (GO:0005515) were significantly enriched (Supplementary Table S18, Supplementary Fig. S8). Additionally, thirteen protein classes were identified, including C2H2 zinc finger transcription factors (PC00248) and RNA metabolism proteins (PC00031). (Supplementary Table S19, Supplementary Fig. S9). No enrichment of hypomethylated genes was identified in pathways. Similarly, a lack of significant enrichment was associated with hypermethylated genes.

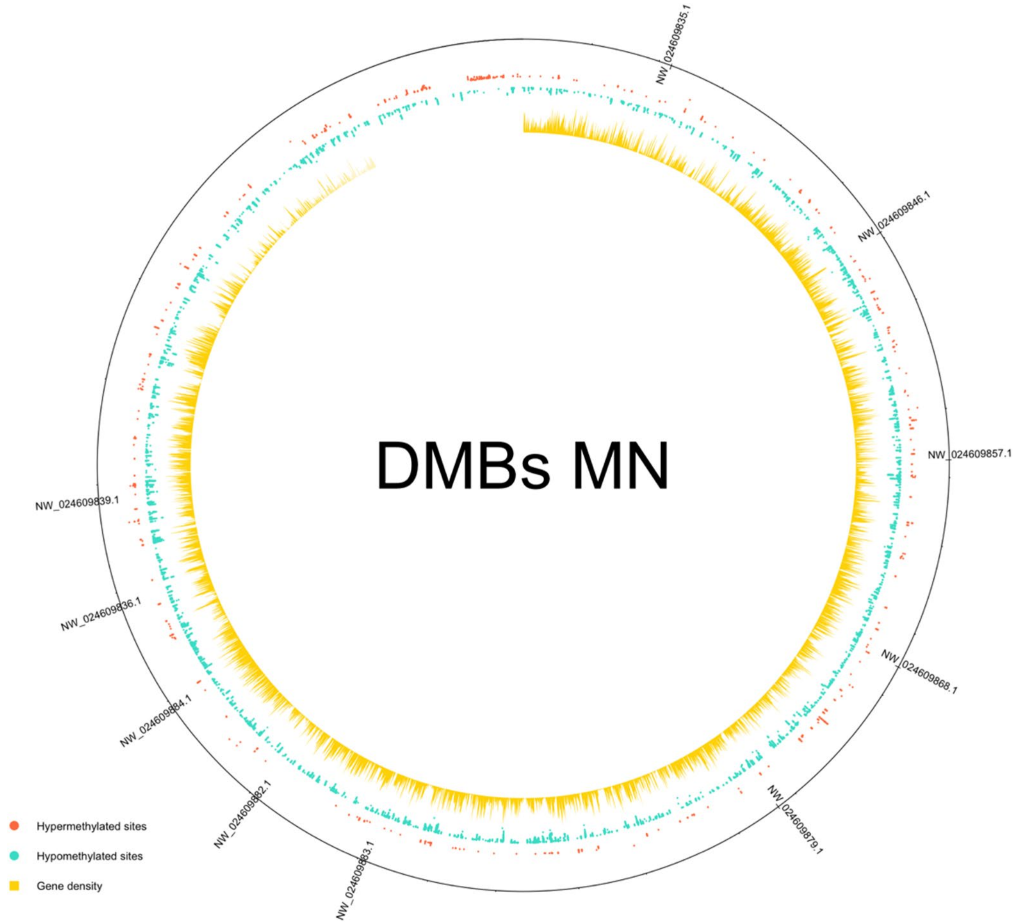

Fig. 6. Distribution of differentially methylated sites in the blacklegged tick genome. The scaffolds with the biggest concentration of differentially methylated bases (DMBs) are labeled within the map. The salmon dots represent hypermethylated bases, the turquoise points indicate hypomethylated bases and yellow peaks display gene density. The abbreviation MN: Minnesota and DMBs: differentially methylated bases.

Discussion

Ixodes scapularis is the most prominent vector of human pathogens in the US, and as such, understanding the molecular mechanisms they use to regulate gene expression and adapt to new environments is of utmost importance. Our study investigated variation in DNA methylation between two blacklegged tick populations in the US, representing the two genetically distinct clades found in the US (MN: All American clade and TX: Southern clade). These two populations presented variable levels of global methylation and differentially methylated genic regions across their genome. Interestingly, assessment of variation in the expression of genes encoding enzymes involved in the homeostasis of DNA methylation revealed that only DNMT1 showed significantly distinct levels of expression, while the expression of the other enzymes was highly variable. Furthermore, our results point at a potential correlation between geographical origin, climatic conditions, and variations in DNA methylation levels.

Using multiple molecular approaches, we discovered that the level of DNA methylation in blacklegged tick genomes varies according to geographical origin and across years. Initial ELISA assays suggested potential population-level variation in %5-mC, but these findings were not consistent across years. Additional methods produced variable results, with nanopore sequencing supporting higher DNA methylation in the northern population, while WGBS indicated lower methylation in MN compared to TX ticks. These discrepancies suggest that DNA methylation in the blacklegged tick may be influenced by factors other than, or in addition to, geographical origin. For instance, unusual meteorological events, such as the historical snowstorm in southern TX in February 2021[45,46], may have altered DNMT relative expression, thereby modifying methylation levels in that blacklegged tick population. Rare climatic events may therefore contribute to population-specific methylation patterns. Thus, interannual climatic events must be considered alongside technical limitations. In addition, technical artifacts, such as low mapping efficiency achieved by our WGBS reads (25% for MN and 31% for TX), may also impact the completeness and representativeness of DNA methylation data. Bisulfite treatment results in lower mapping efficiency, especially when bisulfate conversion leads to mismatches in the first 25 bases at the 5' end[47]. For example, the WGBS of *Haemaphysalis longicornis* resulted in mapping efficiencies between 38.56% – 40.46%[40,48]. Ultimately, this may affect the detection of areas with differentially methylated regions.

Furthermore, technical variability in sequencing technologies and sample heterogeneity could contribute to the disagreement observed across methods. For example, WGBS sequencing was performed on pools of males or females, while nanopore sequencing was used on a single female. As such, our WGBS data presents consensus methylated regions between sexes and multiple individual ticks, which might eliminate regions present in nanopore sequencing. These DMRs detected by nanopore sequencing may be individual-specific and thus not detected in other ticks. Additionally, bisulfite sequencing, while efficient and accurate, cannot distinguish between 5-hmC and 5-mC, potentially overestimating methylated cytosine levels[49,50]. Overall, our results indicate that factors such as the year of sample collection, the choice of methodology, and sample heterogeneity may affect levels of methylation detected. Thus, they must be carefully considered when interpreting these results.

Only three differentially methylated genes were consistently identified as hypermethylated in TX ticks by both sequencing methods: basic proline-rich protein, zinc finger protein 501-like, and an uncharacterized protein LOC115333191. These genes may represent important loci associated with regional adaptation of blacklegged tick to different environments. For instance, the basic proline-rich gene (XM_029973219.3) facilitates the breakdown of food during chewing as a component of mammalian saliva, among other functions[51,52]. Interestingly, this protein presents homology with a zinc finger protein in humans, ZNF608 (NP_001372550.1), which is also present in insects. Curiously, this transcriptional regulator is involved in sexual dimorphism in the broad-horned flour beetle (*Gnatocerus cornutus*)[53]. A second gene encoding a zinc finger protein 501-like (XM_040221513.2), whose human homolog is zinc finger protein 420 (ZNF420), and is involved in the negative regulation of p53-mediated apoptosis under stress conditions[38]. Zinc finger proteins are commonly known as TFs that play crucial roles by regulating gene expression at specific time points. TFs recognize motifs upstream of transcription start sites (TSS) and either activate or inhibit transcription of specific genes, depending on the cellular context[54]. Evidence suggests that the absence of methylation marks in exonic regions can interfere with the recognition of alternative exons. This can influence the generation of alternative transcripts, which may translate into distinct phenotypes[55,56]. Several hypomethylated zinc finger genes were identified in MN ticks, suggesting its dynamic activity may be important for tick epigenetic regulation (Supplementary Table S11, S16). However, the third differentially methylated gene (DMG), uncharacterized protein LOC115333191 (XM_040221333.3), does not have any human homolog, limiting the interpretability of this DMG. Collectively, the WGBS dataset revealed additional hypomethylated genes that may be associated with regional adaptation. These genes included those encoding a serine/threonine-protein kinase pim-1-like (XM_040208937.1), a metalloprotease, cubilin-like (XM_042292858.1), and proline-rich protein (XM_040216711.3), which may serve as a cryoprotectant in insects[57] and plants[58], and a mitogen-activated kinase (XM_029987871.4). It is important to note that this study is correlative, and while these genes may possess biological relevance in tick local adaptation, their specific functional roles in tick biology is currently unknown. Thus, any causal relationship with hypermethylation remains speculative. Future functional studies, such as gene knockdown or epigenetic manipulation, will be essential to confirm the roles of these DMRs and DNMT1 in blacklegged tick adaptation.

Expanding beyond differentially methylated genes, our results also provide evidence that the blacklegged tick genome contains a complete set of genes encoding enzymes that maintain and catalyze de-novo DNA methylation, as well as a complete suite of predicted demethylases (TET and TDG). This comprehensive set of homologs suggests that DNA methylation is a coordinated and dynamic process in blacklegged ticks. While the presence of DNA methylation establishment (DNMT3) and maintenance (DNMT1) enzymes had previously been reported in hard ticks, including blacklegged ticks[31,40,59–61], demethylases had been ignored. Our results indicate that ticks can regulate methylation levels using at least two different mechanisms: (1) addition of methyl groups into cytosine residues to maintain or generate new methylated sites, and (2) removal of methyl groups in response to stimuli or for homeostasis. To this end, environmental stimuli, such as extreme temperatures, has been shown to alter DNMT expression[33,34]. Blacklegged ticks experience considerable environmental variation, likely requiring unique and rapid molecular responses. For instance, in other hard tick species, transient cold exposure increases DNMT expression[60], suggesting that this enzyme may play a role in winter survival. Additionally, DNA methylation may be involved in development, as prior research found variation of methylation levels in different life stages of *Ixodes* species[62]. However, the specific role of DNMTs, particularly cytosine modifying enzymes, are beyond the scope of this study. Further research is required to clarify the function of DNA methylation in blacklegged ticks.

Notably, in addition to identifying the components of the methylation machinery, we found significant variation in DNMT1 relative expression (Supplementary Fig. S3). This indicates that blacklegged tick populations may differ in their ability to maintain DNA methylation profiles. Sexual dimorphism is apparent in blacklegged ticks, where females are generally larger and possess a smaller scutum than males[11]. Blacklegged ticks, like humans, contain XY chromosomes as males and XX chromosomes as females[35]. Yet, the potential molecular mechanism(s) contributing to sexual dimorphism in *I. scapularis* has not been identified. Interestingly, TX females exhibited higher relative expression of DNMT1 than males of the same location (Fig. 3). Intersex variation in DNMT1 expression levels has been recorded in other arthropods, such as the mealybug (Hemiptera: Pseudococcidae). In this case, mealybug females showed higher DNMT1 expression levels than males, suggesting that enzymes involved in the maintenance of DNA methylation may contribute to sex-specific methylation patterns and sexual differentiation[63]. Nevertheless, the present study is correlational and does not provide direct evidence that supports the role of DNA methylation in sexual differentiation. Further studies are needed to assess the role of IscDNMT1 and DNA methylation in blacklegged tick sexual differentiation.

The expression of the remaining enzymes showed considerable variation in relative expression levels without any consistent trend, which might reflect the wide range of global 5-mC levels uncovered in blacklegged tick genomes. Nevertheless, global 5-mC methylation in blacklegged tick genomes was consistent with levels reported in other chelicerate species, including other ticks[31,40]. Nwanade et al.[40] reported approximately 3% 5-mC levels in CpG contexts in *H. longicornis* females from a laboratory colony. However, laboratory-reared ticks raised under controlled settings might not accurately reflect the suite of abiotic and biotic stressors shaping DNA methylation in natural tick populations. Epigenetic modifications in cis-regulatory elements, such as promoters, coordinate the spatiotemporal transcriptional programs in arthropods[64]. Still, how this methylation contributes to modulating gene expression in blacklegged ticks remains unknown. We identified CpG sites that were hyper- and hypomethylated in MN ticks when compared to TX ticks, falling within 78 predicted promoter regions. For instance, possible promoter regions of TFs in ticks, such as zinc finger genes[35] were found to be either highly or lowly methylated between our tick populations. This differential methylation could ultimately affect their regulatory activity, either by blocking transcription factor binding or enhancing gene transcription[54]. A second mechanism by which DNA methylation may affect gene regulation is through CpG islands and island shores, which are stretches of DNA with high densities of CpG bases. These islands can be methylated, leading to gene silencing, especially during imprinting[65]. Additionally, the upstream regions of genes encoding transposable elements (TEs) were found to be hypomethylated in northern compared to southern tick populations; these regions are highly repetitive and abundant in the *Ixodes* genome[35]. Although the precise effects of TEs in *Ixodes* ticks are still under investigation[66] and remain unclear, studies in insects suggest that these elements may provide benefits that enhance adaptation to shifting environmental conditions via exaptation[67]. Nonetheless, whether methylation of promoter regions or CpG islands in ticks results in inhibition or induction of gene expression remains to be determined.

Finally, variations in DNA methylation, either depletion or addition, have been recorded as a response to distinct climatic conditions experienced in unique geographical locations by different populations of the same species[44,68]. To our knowledge, our study provides the first report on the relative expression levels of each enzyme involved in DNA methylation homeostasis, quantitative characterization of global DNA methylation levels, and computation of differentially methylated genomic regions among blacklegged tick populations from contrasting geographic regions in the US. In the absence or presence of genetic variation, epigenetic mechanisms can contribute dynamically to phenotypic plasticity in response to environmental stimuli[69–72]. Given that *Ixodes* ticks have a complex life cycle with prolonged off-host periods that expose them to several environmental stressors, ticks are under variable selective pressures to respond to external stimuli. Environmental stimuli can demonstrably shape epigenetic responses, particularly by modulating DNA methylation profiles. External factors, such as transitions between freshwater and saltwater[68], changes in salinity[73], exposure to novel habitats[74], or pesticides[75], can induce alterations in DNA methylation with functional relevance to physiological responses. Here, environmental stimuli refers to any external changes in environmental conditions that can modulate DNA methylation profiles in individuals[74]. However, it remains unclear which specific external cues can alter the epigenetic landscape of organisms[76]. In hard ticks, factors such as high temperatures[33,77], host preference[78,79], photoperiod[42], and winter temperatures[40,44] may contribute to variation in interpopulation methylation. Future studies are needed to clarify the relationship between environmental factors and epigenetic variation in this tick species.

Contrasting environmental conditions experienced by northern and southern populations of blacklegged ticks can result in genetically distinct groups that develop region-specific strategies. Building on this idea, previous studies have demonstrated marked genetic differentiation along the north-south divide, revealing evidence of local adaptation, particularly at loci associated with environmental conditions and host availability[43,80]. In this context, host preference emerges as a crucial factor influencing pathogen transmission[27]. Despite lower overall genetic differentiation, northern ticks possess greater variations in host-interaction genes than their southern counterparts[80]. This genetic diversity may enable northern ticks to feed on a wider variety of hosts, including *Peromyscus* species, the primary reservoir of *B. burgdorferi* in the US[26,81]. In contrast, ground-dwelling hosts such as lizards, which are common hosts for blacklegged ticks in the south, are poor reservoirs for the pathogen[26,82]. As a result, host preference in the north increases contact with competent mammalian hosts, increasing the likelihood of pathogen transmission to humans[26]. Extending these findings, variation in DNA methylation among northern and southern tick populations might contribute to local adaptation and, in turn, influence the epidemiology of tick-borne pathogens in North America. Geographical haplotypes in other tick species have shown differences in vector competence[83]. Yet, there is currently no evidence supporting variation in vector competence among blacklegged tick clades.

Overall, this work provides an initial resource for understanding the impact of epigenetic modifications in tick biology. However, a direct causal link to activity or vectorial capacity was not established in this study. Future research should experimentally evaluate the causal relationships between environmental drivers, methylation changes, and disease transmission, with the goal of informing targeted strategies to mitigate tick-borne diseases.

Methods

Tick collections

Questing adult blacklegged tick were collected from distinct locations in MN and TX (Supplementary Table S20) between 2016 and 2023, except for 2018 and 2020 when collection was not performed. Females and males were collected from the vegetation along trails with tick drags (BioQuip, Rancho Dominguez, CA, US) and placed in 100% ethanol or RNAlater (Invitrogen, Waltham, MA, US). Only unfed adults were used for these experiments. Immature *I. scapularis* stages are rarely collected from the vegetation in the south[84,85] and, we did not collect any during our study period. Moreover, methylation from the bloodmeal within engorged ticks may affect our results and maintaining ticks at laboratory conditions until molting may also affect their methylation patterns. Therefore, we decided to perform these analyses on unfed adult ticks collected from the field only. While we recognized that pathogens might exert epigenetic control[86–88], the influence of pathogen infections on DNA methylation in ticks remains unclear and warrants further investigation. Collected ticks were immediately stored at -80 °C until DNA and RNA isolation was carried out as described below.

Quantification of 5-methylcystosine (5-mC) levels in blacklegged tick

Ticks were defrosted and removed from the storage solution and washed three times with 1 ml 1x Phosphate Buffered Saline (PBS). They were then separated by sex into different groups, depending on the downstream procedure as described below. Preliminary measurements of 5-mC levels were performed using DNA from ticks collected in Chippewa National Forest, Camp Ripley and Saint Croix State Park, MN and Harris County, TX in 2017 (Supplementary Table S20). Methylation levels were measured with MethylFlash Global DNA Methylation (5-mC) ELISA Easy Kit [Colorimetric] (Epigentek, Farmingdale, NY, US).

All following assays were performed using adult ticks collected from 2021 to 2022 from MN and TX (Supplementary Table S20). To ensure adequate statistical power, DNA methylation was quantified from at least five individuals of each sex at each study location. Samples were distributed across four independent experimental replicates for analysis. DNA was isolated using the Quick-DNA/RNA Microprep Plus kit (Zymo Research Corporation) according to the manufacturer indications with the following modifications: 200 µL of the lysis buffer was used to homogenize the ticks using a Fisherbrand RNase-free disposable pestle (Fisher Scientific, Waltham, MA, US) and 30 µL of DNA/RNase free water for the final elution of DNA. DNA concentration and integrity were assessed with a NanoQuant Infinite M200PRO (Tecan Group Ltd., Mannedorf, Switzerland) with the i-control 1.12 software and stored at -80 °C until use. The DNA cytosine methylation in the blacklegged tick was measured using the 5-mC ELISA kit (Zymo Research Corporation), according to the manufacturer's instructions. In brief, 100 ng of DNA were denatured at 98 °C for 5 min and applied to each well. 5-mC was detected using a mouse anti-5-methylcytosine monoclonal antibody (clone 7D21; Zymo Research Corporation) and an HRP-labeled secondary antibody. Samples were assessed in duplicates. Each assay included a standard curve with known 5-mC percentages to determine the methylation levels in the samples. Change in color, which reflects the presence of 5-mC, was quantified using a NanoQuant Infinite M200PRO (Tecan Group Ltd.) with Magellan 7.1 with an absorbance of 450 nm. The %5-mC was calculated using a standard curve. Statistical differences in the 5-mC level were determined using an unpaired two-tailed t-test in GraphPad Prism 9.2.0 (GraphPad Software, San Diego, CA, US). Differences in 5-mC levels were compared using geographic location and sex as variables.

Characterization of 5-mC patterns in texas and minnesota blacklegged tick populations using WGBS

The contrast in the differences in global levels of methylation and the methylation of specific regions was assessed by whole genome bisulfite sequencing (WGBS). DNA was isolated from three groups of pooled ticks (five to six ticks), separated by sex and location, using the Quick-DNA/RNA Microprep Plus kit (Zymo Research) following the manufacturer directions with a few modifications. Briefly, ticks were crushed in 400 µL of lysis buffer using Fisherbrand RNase-free disposable pestles (Fisher Scientific) and eluted with a final volume of 50 µL of DNA/RNase free water per sample. DNA from three groups of female and male ticks were submitted to GENEWIZ (South Plainfield, NJ, US) on dry ice for WGBS. DNA quality was assessed in a Qubit 2.0 Fluorometer (Life Technologies, Carlsbad, CA, US) to detect DNA concentration and possible RNA contamination. Samples with RNA contamination were RNAse treated before building libraries.

Bisulfite conversion, library preparation and sequencing were also conducted byGENEWIZ. In brief, Bisulfite treatment was performed with the EZ DNA Methylation Gold Kit (Zymo Research). DNA was fragmented with Covaris (PerKinElmer Covaris, Woburn, MA, US) and library preparation was performed using the Accel-NGS Methyl-Seq kit (Swift Biosciences, Ann Arbor, MI, US). A ~ 0.5% unmethylated lambda DNA spike-in was used as a bisulfite conversion rate control and for downstream bioinformatics. Sequencing libraries were validated using the Agilent Tapestation 4200 (Agilent Technologies, Palo Alto, CA, US) and quantified by Qubit 2.0 fluorometer as well as by quantitative real-time PCR (Applied Biosystem, Carlsbad, CA, US). Finally, sequencing libraries were multiplexed and sequenced on the Illumina HiSeq instrument (4000 or equivalent) according to manufacturer's instructions. The samples were sequenced using a 2 × 150 paired-end (PE) configuration. The information and accession number for each sequence is provided in Supplementary Table S21.

To identify changes in DNA methylation sites between populations, WGBS data were quality filtered and trimmed using Trim Galore v0.6.10 (--paired -q 30 --illumina --length 35 --clip_R1 10 --clip_R2 10 --three_prime_clip_R1 5 --three_prime_clip_R2 5)[89]. Filtered bisulfite reads were mapped to the reference genome (NCBI, ASM1692078v2;[35] using bowtie2 v2.5.4. This assembly was selected for its high contiguity and dense gene annotation, both of which are essential for robust DMR detection. Duplicates were then removed and methylated cytosines called with Bismark v0.24.2[90]. Hyper/hypomethylated sites were processed by methylKit package v1.36.0[91] implemented in RStudio v2026.1.0.392[92] with a q-value cutoff at 0.01 and the minimum difference in the methylation levels was set to 25% value. DMBs were compared with gene annotation (NCBI, ASM1692078v2). Genes with at least one hypo or hypermethylated site were considered as a hyper- or hypomethylated gene. methylKit and circlize v0.4.17 packages[93] were used to visualize and map chromosome-wide differentially methylated region distribution and associated with feature annotations in the WGBS datasets.

To ensure that observed variability in methylation between the MN and TX blacklegged tick populations were due to epigenetic rather than genetic variation. WGS sequences were obtained from two biological replicates per population collected in 2021(Supplementary Table S21).These sequences were used to filter out variable sites that overlap methylated cytosine in CpG context, following the Rahman and Lozier approach[44].

DNA was extracted from single ticks and prepared for Illumina sequencing. Individuals were homogenized in 1.5 mL tubes in liquid nitrogen baths with liquid nitrogen-cooled pestles. DNA was isolated using the E.Z.N.A. Insect DNA Kit (Omega Bio-tek, Inc., Norcross, GA, US). Following isolation, genomic DNA was evaluated using a NanoDrop 2000 Spectrophotometer (Thermo Fisher Scientific, Waltham, MA, US). Illumina libraries were prepared using the FS DNA Library Prep Kit and NEBNext Multiplex Oligos for Illumina (New England Biolabs, Ipswich, MA, US). Size selection and cleanup were performed with SPRIselect beads (Beckman

Coulter, Inc., Brea, CA, US). The prepared libraries were sequenced at the USDA-ARS Veterinary Pest Genetics Research Unit in Kerrville, Texas, on a NextSeq 2000 system with a P3 flow cell and a 200-cycle reagent cartridge (Illumina, Inc., San Diego, CA, US).

To identify variable sites such as SNPs, raw sequences were trimmed and filtered out using bbduk v39.11[94]. Trimmed reads were subsequently mapped to the *I. scapularis* reference genome (NCBI, ASM1692078v2) using the BWA-MEM algorithm from BWA v07.17[95]. The sequence alignment map (SAM) files were then converted to binary format and sorted by coordinates using SAMtools v1.19.2[96]. Mark duplication and the index of binary alignment map (BAM) files were conducted with Picard v3.2.0[97]. The output files were used for SNP identification with FreeBayes v1.3.8[98]. To simplify data interpretation and mitigate potential sequencing artifacts that could introduce errors downstream[99,100], the initial variable site set was further refined using VCFtools v0.1.16[101]. This process involved removing indels, non-biallelic SNPs, sites with low read depth, quality score $Q \geq 20$, and minor allele counts ≥ 2. Additionally, SNPs with excessive coverage were excluded from the final VCF file, utilizing VCFtools along with AWK v5.1.0 (https://www.gnu.org/software/gawk/manual/gawk.html#Manual-History), as outlined by Rahman and Lozier[44][https://github.com/steph166/contrastingEpigenetics-].

SNPs were compared to hyper- and hypomethylated genes using the GenomicRanges package[102] in R v4.5.2 and VLOOKUP function in Excel. Any differentially methylated genes overlapping with the SNP site were excluded from the final list. Final lists of hyper- and hypomethylated genes were generated by removing any gene that contained both hypo- and hypermethylated sites, as well as those overlapping with their counterpart list, based on nanopore sequencing data.

Additionally, predicted promoter regions in blacklegged tick were analyzed for variation in 5-mC levels, which might influence gene expression[103,104]. Thus, DMBs were examined in search of hyper- and hypomethylated promoter regions. Promoter regions in this tick species were defined as those located within 2000 bp upstream of the TSS[105–107]. A new column titled “promoter” was incorporated into the gene information set, containing the beginning of each potential promoter region and the end of the potential region, with gene directionality considered to accurately delimitate the defined promoter regions. Subsequently, differentially methylated genes previously identified were merged with gene information (GenesLocations), using “scaffold name” as a common column between the two datasets. Candidate differentially methylated promoter regions were then identified based on the coordinates of the putative promoter region, and the gene start site (see Supplementary Table S15 for the formulas). The final dataset was generated by excluding any SNPs that overlapped with differentially methylated sites identified in this analysis. This dataset was used to assess gene ontology or pathway enrichment as described below.

Gene ontology associated with hyper- and hypomethylated genes in blacklegged tick populations

Gene ontology and pathway analyses were conducted for hyper- and hypomethylated genes identified in MN ticks in comparison to TX ticks, using the Panther Classification System v19.0[108,109]. The statistical overrepresentation test was selected for the analysis, with Fisher’s test employed as the default statistical method, and visualization was performed using ggplot2 v4.0.2[110] in RStudio v2026.1.0.392[92].

DNA isolation and nanopore sequencing

To determine whether the variations in DNA methylation observed with WGBS could be detected using another approach, *I. scapularis* DNA was sequenced using nanopore. Blacklegged ticks collected from Carlos Avery Wildlife Management Area, MN and Beech woods trail, Big Thicket State Park, TX in 2023[80](Supplementary Table S21) were morphologically identified using keys from Kierans and Clifford[111] and Cooley and Kohls[112]. Ticks were rinsed with molecular grade water and transferred to DNA/RNA shield until DNA extraction. DNA was extracted using a Qiagen MagAttract kit (Qiagen, Germantown, MD, US) according to the manufacturer’s instructions, with a final elution step extended to 1 h at 37 °C. DNA was sheared by 20 passes through a standard 30-gauge insulin syringe. Library prep was performed with an SQK-LSK-114 native ligation sequencing kit (ONT, Oxford, United Kingdom). Sequencing was performed on a P2 Solo instrument and base called using Nvidia 4090 GPUs. Sequencing was performed over three days with nuclease flush and library reload every 24 h. Two adult females were used in total for sequencing of blacklegged ticks, one from MN and one from TX.

Characterization of 5-mC patterns in texas and minnesota *I. scapularis* populations using oxford nanopore technologies

Nanopore sequencing is a single-molecule sequencing platform that leverages electrical currents across nanopores embedded in a flow cell to detect individual nucleotides of a DNA fragment as they pass through the nanopores. The change in electrical current is determined by the molecular weight of the passing nucleotide, which is deciphered through a bioinformatic process known as base calling. Methylated bases possess unique molecular weights and can be detected natively. We implemented nanopore sequencing as an alternative method to validate the global 5-mC methylation levels and DMRs observed with WGBS. Raw data from all runs were base called together using Dorado v0.8.1 with the “super accuracy” model dna_r10.4.1_e8.2_400bps_supv4.2.0. Base modifications were called simultaneously using the Dorado flag —modified-bases 5mC_5hmC. Read quality was assessed using Nanoq v0.10.0[113]. Global DNA methylation and hydroxymethylation at cytosine-guanine dinucleotides (CpGs) were identified using modified base information stored in the initial base calling output files for two female individual blacklegged ticks (MN and TX, respectively)[114]. The two blacklegged tick individual output files were concatenated and aligned to the PalLabHiFi assembly using Minimap2 v2.28[115]. The resulting mapped BAM files containing modified base information were converted to bedMethyl format using Modkit v0.4.1[116], and global 5-mC and 5hmC percentages were calculated using AWK v5.3.

Differential methylation analyses were performed according to Flack et al.[114]. Count filtering, tiling into 100 bp windows, and differential methylation analysis were performed with methylKit v1.30.0[91]. Windows were considered DMRs if they had at least 10 CpGs, a mean absolute difference in methylation ≥ 50%, and a Benjamini–Hochberg-adjusted P-value of 0.05[117]. CpG sites are locations on a DNA strand where a cytosine is directly followed by a guanine nucleotide, connected by a phosphate bond. The cytosine within the CpG dinucleotide context can be methylated. DMRs were annotated by identifying the nearest gene with BedTools[118]. DMRs identified between individuals were visualized with karyotypeR v1.30.0[119].

Identification of demethylation and methylation enzymes homologous in blacklegged tick populations

DNA methylation is carried out by a group of enzymes that add a methyl group onto the 5th carbon in cytosine, DNMTs, which are traditionally named DNMT1, DNMT2, and DNMT3[65,120]. Demethylation in the genome is regulated by TET dioxygenases and TDG[104]. To establish whether these enzymes are present in the genome of the blacklegged tick, we used the protein sequences of *Homo sapiens* DNMT1 (AAI26228.1), DNMT2 (CAG29312.1), DNMT3a (AAH23612.1), and TET3 (NP_001274420.1) to identify homologs. Additionally, *Mus musculus* TDG (NP_766140.2) was used. Homologs were determined by Position-Specific Iterative (PSI)-BLAST search (NCBI, NIH, Bethesda, MD, US). The identity of the homologs was corroborated by confirming the presence of conserved domains within the proteins, using SMART v9[39].

The Expasy Translate Tool (Swiss Institute of Bioinformatics, Lausanne, Switzerland;[121] was used to discard any non-protein coding sequence and identify the mRNA coding sequence for each enzyme, except TET3, for which several isoforms were identified. Trimmed sequences were used to design oligos using the Geneious Prime v2021.2 (Biomatters Inc, Auckland, New Zealand) and OligoAnalyzer Tool in IDT (Integrated DNA Technology Inc, Coralville, IA, US;[122]. In the case of DNMT3 and TDG, probes were designed using the PrimerQuest Tool. In the case of TET3, all isoform sequences were aligned in Geneious Prime v.2021.2 to locate a conserved region among all isoforms and primers were designed using this region (see Supplementary Fig. S10).

To ascertain the expression of these enzymes in both populations of blacklegged ticks, RNA from individual ticks was purified from the lysate flow-through of the purified DNA from the ticks used in our 5-mC quantification experiments described above. RNA was isolated using the Quick-DNA/RNA Microprep Plus kit (Zymo Research), according to the manufacturer's indications. RNA was eluted with 30 µl of DNA/RNase free water. cDNA was synthesized from the mRNA isolated from ticks collected from either MN or TX, using the Verso cDNA Synthesis Kit (Thermo Scientific), following the manufacturer's protocols. Amplicons were amplified using GoTaq Flexi DNA Polymerase (Promega Corporation, Madison, WI) with the following thermal cycling conditions: one cycle of 95 °C for 3 min, followed by 95 °C for 60 s, 52–55 °C for 60 s, and 72 °C for 30 s per 34 cycles, and a final extension at 72 °C for 5 min. To verify the identity of the amplicons, PCR products were purified using QIAquick PCR purification kit (Qiagen), according to the manufacturer's instructions. Purified PCR products were submitted to EtonBio (Eton Bioscience, Inc., San Diego, CA, US) for Sanger sequencing. The sequence identity was confirmed by BLAST and deposited in NCBI (Supplementary Table S22).

Measurement of DNA methylation and demethylation enzymes relative expression

In other arthropod species, such as bees and ants, stimuli and memory formation can lead to the differential expression of DNMTs and TET[123,124]. Likewise, in ticks, cold temperatures can affect the expression of DNMTs[60]. To assess whether methylating and demethylating enzymes were differentially expressed in blacklegged tick populations, the primers listed in Supplementary Table S23 were used to assess the relative expression of DNMTs, TET, and TDG enzymes. The mRNA was normalized to 100 ng and used for cDNA synthesis as described above.

The relative expression of DNMT1, DNMT2, and TET was evaluated using SyBR Green (Thermo Fisher Scientific, Carlsbad, CA, US), according to the manufacturer's directions. However, DNMT3 and TDG were measured using TaqMan Fast Advanced Master Mix (Thermo Fisher Scientific), due to the presence of primer dimer and secondary products that could not be eliminated. All primers and probes are listed in Supplementary Table S23. All denaturing steps were performed at 95 °C for 60 s and annealing temperature between 52 and 55 °C for 60 s over 39 cycles (see Supplementary Table S23, S24 for details about reaction conditions). Fluorescence was recorded during the annealing step to evaluate amplification. The specificity of the primers was verified by adding a melting curve step in the SyBR Green assays. Three biological replicates were performed to evaluate the relative expression of each gene. For the DNMT1 gene, four replicates were performed to achieve a consensus. All reactions were performed in a CXF Opus 96 (Bio-Rad Laboratories Inc, Hercules, CA, US).

The relative expression of methylation and demethylation enzymes was calculated using $2^{-\Delta CT}$ method[125], using actin as a normalizing gene[126]. Significant differences in gene expression were evaluated using a two-way ANOVA in GraphPad Prism 9.4.1 (GraphPad Software, San Diego, CA, US).

Data availability

Raw data from WGBS and WGS associated with this manuscript has been deposited in the National Center for Biotechnology Information (NCBI) under Bioproject number PRJNA1081399, and the nanopore raw reads can be found at https://datadryad.org/dataset/doi:10.5061/dryad.sbcc2frh8#citations. All code used for the analysis of the data, along with the raw output, is available in the public repository: https://github.com/steph166/contrastingEpigenetics-.Accession numbers, PCR and qPCR amplification conditions, WGBS, WGB, and nanopore statistics and results, as well as a full list of differentially methylated sites and enrichment analyses in putative promoters and genes, are available in the supplementary material associated with this manuscript.

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References

1. Centers for Disease Control and Prevention (CDC). Tickborne disease surveillance data summary. Database: CDC Ticks (2024). https://www.cdc.gov/ticks/data-research/facts-stats/tickborne-disease-surveillance-data-summary.html (2024).
2. Kugeler, K. J., Schwartz, A. M., Delorey, M. J., Mead, P. S. & Hinckley, A. F. Estimating the frequency of Lyme disease diagnoses, United States, 2010–2018. *Emerg. Infect. Dis.* **27**, 616. https://doi.org/10.3201/eid2702.202731 (2021).
3. Rochlin, I. & Toledo, A. Emerging tick-borne pathogens of public health importance: a mini-review. *J. Med. Microbiol.* **69**, 781–791 (2020).
4. Hook, S. A. et al. Economic burden of reported Lyme disease in high-incidence areas, United States, 2014–2016. *Emerg. Infect. Dis.* **28**, 1170. https://doi.org/10.3201/eid2806.211335 (2022).
5. Steiner, F. E. et al. Infection and co-infection rates of Anaplasma phagocytophilum variants, Babesia spp., Borrelia burgdorferi, and the rickettsial endosymbiont in Ixodes scapularis (Acari: Ixodidae) from sites in Indiana, Maine, Pennsylvania, and Wisconsin. *J. Med. Entomol.* **45**, 289–297 (2014).
6. Eisen, R. J. & Eisen, L. The blacklegged tick, Ixodes scapularis: an increasing public health concern. *Trends Parasitol.* **34**, 295–309 (2018).
7. Prusinski, M. et al. Prevalence of *Borrelia burgdorferi* (Spirochaetales: Spirochaetaceae), *Anaplasma phagocytophilum* (Rickettsiales: Anaplasmataceae), and *Babesia microti* (Piroplasmida: Babesiidae) in *Ixodes scapularis* (acari: Ixodidae) collected from recreational lands in the Hudson Valley region, New York state. *J. Med. Entomol.* **51**, 226–236 (2014).
8. Lehane, A. et al. Prevalence of single and coinfections of human pathogens in Ixodes ticks from five geographical regions in the United States, 2013–2019. *Ticks Tick. Borne Dis.* **12**, 101637. https://doi.org/10.1016/j.ttbdis.2020.101637 (2021).
9. Hersh, M. H. et al. Co-infection of blacklegged ticks with Babesia microti and Borrelia burgdorferi is higher than expected and acquired from small mammal hosts. *PLoS One*. **9**, e99348. https://doi.org/10.1371/journal.pone.0099348 (2014).
10. Dye-Braumuller, K. C. et al. Riding the wave: reactive vector-borne disease policy renders the United States vulnerable to outbreaks and insecticide resistance. *J. Med. Entomol.* **59**, 401–411 (2022).
11. Keirans, J. E., Hutcheson, H. J., Durden, L. A. & Klompen, J. *Ixodes (Ixodes) scapularis* (Acari: Ixodidae): redescription of all active stages, distribution, hosts, geographical variation, and medical and veterinary importance. *J. Med. Entomol.* **33**, 297–318 (1996).
12. Eisen, R. J., Eisen, L. & Beard, C. B. County-scale distribution of *Ixodes scapularis* and *Ixodes pacificus* (Acari: Ixodidae) in the continental United States. *J. Med. Entomol.* **53**, 349–386 (2016).
13. Eisen, L. & Eisen, R. J. Changes in the geographic distribution of the blacklegged tick, *Ixodes scapularis*, in the United States. *Ticks Tick. Borne Dis.* **14**, 102233. https://doi.org/10.1016/j.ttbdis.2023.102233 (2023).
14. Nieto, N. C. & Foley, J. E. Meta-analysis of coinfection and coexposure with *Borrelia burgdorferi* and *Anaplasma phagocytophilum* in humans, domestic animals, wildlife, and *Ixodes ricinus*-complex ticks. *Vector-Borne Zoonotic Dis.* **9**, 93–102 (2009).
15. Paulson, A. R., Lougheed, S. C., Huang, D. & Colautti, R. I. Multiomics reveals symbionts, pathogens, and tissue-specific microbiome of blacklegged ticks (*Ixodes scapularis*) from a Lyme disease hot spot in southeastern Ontario, Canada. *Microbiol. Spectr.* **11**, e0140423. https://doi.org/10.1128/spectrum.01404-23 (2023).
16. Centers for Disease Control and Prevention (CDC). Lyme disease surveillance data. Database: CDC Lyme disease (2025). https://www.cdc.gov/lyme/data-research/facts-stats/surveillance-data-1.html (2025).
17. 17 Xu, G., Wielstra, B. & Rich, S. M. Northern and southern blacklegged (deer) ticks are genetically distinct with different histories and Lyme spirochete infection rates. *Sci. Rep.* **10**, 10289. https://doi.org/10.1038/s41598-020-67259-0 (2020).
18. Hamer, S. A., Hickling, G. J., Walker, E. D. & Tsao, J. I. Increased diversity of zoonotic pathogens and *Borrelia burgdorferi* strains in established versus incipient *Ixodes scapularis* populations across the Midwestern United States. *Infect. Genet. Evol.* **27**, 531–542 (2014).
19. Frederick, J. C. et al. Phylogeography of the blacklegged tick (*Ixodes scapularis*) throughout the USA identifies candidate loci for differences in vectorial capacity. *Mol. Ecol.* **32**, 3133–3149 (2023).
20. Van Zee, J. et al. High SNP density in the blacklegged tick, Ixodes scapularis, the principal vector of Lyme disease spirochetes. *Ticks Tick. Borne Dis.* **4**, 63–71 (2013).
21. Goddard, J., Embers, M., Hojgaard, A. & Piesman, J. Comparison of tick feeding success and vector competence for *Borrelia burgdorferi* among immature *Ixodes scapularis* (Ixodida: Ixodidae) of both southern and northern clades. *J. Med. Entomol.* **52**, 81–85 (2015).
22. Oliver Jr, J. H. et al. Conspecificity of the ticks Ixodes scapularis and I. dammini (Acari: Ixodidae). *J. Med. Entomol.* **30**, 54–63 (1993).
23. Black, W. 4 & Piesman, J. Phylogeny of hard-and soft-tick taxa (Acari: Ixodida) based on mitochondrial 16S rDNA sequences. *Proc. Natl. Acad. Sci.* 91, 10034–10038 (1994).
24. Ogden, N. H. et al. Evidence for geographic variation in life-cycle processes affecting phenology of the Lyme disease vector *Ixodes scapularis* (Acari: Ixodidae) in the United States. *J. Med. Entomol.* **55**, 1386–1401 (2018).
25. Stromdahl, E. & Hickling, G. Beyond Lyme: Aetiology of tick-borne human diseases with emphasis on the South-Eastern United States. *Zoonoses Public. Health*. **59**, 48–64 (2012).
26. Ginsberg, H. S. et al. Why Lyme disease is common in the northern US, but rare in the south: The roles of host choice, host-seeking behavior, and tick density. *PLoS Biol.* **19**, e3001066. https://doi.org/10.1371/journal.pbio.3001066 (2021).
27. Kelly, P. H., Willis, S., Davidson, A., Stark, J. H. & Moïsi, J. C. Estrada-Peña, A. Variables for habitat and vertebrate hosts of Ixodes scapularis are the best ecological predictors of the spatial spread of Lyme disease in the United States (2010–2019). *Parasites Vectors*. **18**, 410 (2025).
28. Arsnoe, I. M., Hickling, G. J., Ginsberg, H. S., McElreath, R. & Tsao, J. I. Different populations of blacklegged tick nymphs exhibit differences in questing behavior that have implications for human Lyme disease risk. *PLoS One*. **10**, e0127450. https://doi.org/10.1371/journal.pone.0127450 (2015).
29. Arsnoe, I., Tsao, J. I. & Hickling, G. J. Nymphal Ixodes scapularis questing behavior explains geographic variation in Lyme borreliosis risk in the eastern United States. *Ticks Tick. Borne Dis.* **10**, 553–563 (2019).
30. Tietjen, M., Esteve-Gasent, M. D., Li, A. Y. & Medina, R. F. A comparative evaluation of northern and southern Ixodes scapularis questing height and hiding behaviour in the USA. *Parasitology* **147**, 1569–1576 (2020).
31. Lewis, S. H. et al. Widespread conservation and lineage-specific diversification of genome-wide DNA methylation patterns across arthropods. *PLoS Genet.* **16**, e1008864. https://doi.org/10.1371/journal.pgen.1008864 (2020).
32. Glastad, K. M., Hunt, B. G. & Goodisman, M. A. Epigenetics in insects: genome regulation and the generation of phenotypic diversity. *Annu. Rev. Entomol.* **64**, 185–203 (2019).
33. Dai, T. M. et al. Molecular characterizations of DNA methyltransferase 3 and its roles in temperature tolerance in the whitefly, *Bemisia tabaci* Mediterranean. *Insect Mol. Biol.* **27**, 123–132 (2018).
34. Wang, Y. et al. J.-d. Molecular characterization of DNA methyltransferase 1 and its role in temperature change of armyworm *Mythimna separata* Walker. *Arch. Insect Biochem. Physiol.* **103**, e21651. https://doi.org/10.1002/arch.21651 (2020).
35. De, S. et al. A high-quality Ixodes scapularis genome advances tick science. *Nat. Genet.* **55**, 301–311 (2023).
36. Glastad, K., Hunt, B. & Goodisman, M. Evidence of a conserved functional role for DNA methylation in termites. *Insect Mol. Biol.* **22**, 143–154 (2013).

37. Norris, D. E., Klompen, J., Keirans, J. E. & Black, I. V. Population genetics of Ixodes scapularis (Acari: Ixodidae) based on mitochondrial 16S and 12S genes. *J. Med. Entomol.* **33**, 78–89 (1996).
38. National Center for Biotechnology Information (NCBI). Database: National Library of Medicine (NIH). NIH (1988). https://www.ncbi.nlm.nih.gov/
39. Letunic, I., Khedkar, S. & Bork, P. SMART: recent updates, new developments and status in 2020. *Nucleic Acids Res.* **49**, D458–D460 (2021).
40. Nwanade, C. F. et al. DNA methylation variation is a possible mechanism in the response of *Haemaphysalis longicornis* to low-temperature stress. *Int. J. Mol. Sci.* **23**, 15207. https://doi.org/10.3390/ijms232315207 (2022).
41. Liu, S., Aageaard, A., Bechsgaard, J. & Bilde, T. DNA Methylation Patterns in the Social Spider, *Stegodyphus dumicola*. *Genes (Basel).* **10**, 137. https://doi.org/10.3390/genes10020137 (2019).
42. Pegoraro, M., Bafna, A., Davies, N. J., Shuker, D. M. & Tauber, E. DNA methylation changes induced by long and short photoperiods in Nasonia. *Genome Res.* **26**, 203–210 (2016).
43. Schoville, S. D. et al. Genome resequencing reveals population divergence and local adaptation of blacklegged ticks in the United States. *Mol. Ecol.* **33**, e17460. https://doi.org/10.1111/mec.17460 (2024).
44. Rahman, S. R. & Lozier, J. D. Genome-wide DNA methylation patterns in bumble bee (*Bombus vosnesenskii*) populations from spatial-environmental range extremes. *Sci. Rep.* **13**, 14901. https://doi.org/10.1038/s41598-023-41896-7 (2023).
45. Bolinger, R. A. et al. An assessment of the extremes and impacts of the February 2021 South-Central U.S. Arctic outbreak, and how climate services can help. *Weather Clim. Extrem.* **36**, 100461 (2022).
46. Veettil, A. V., Fares, A. & Awal, R. Winter storm Uri and temporary drought relief in the western climate divisions of Texas. *Sci. Total Environ.* **835**, 155336 (2022).
47. Porter, J., Sun, M., Xie, H. & Zhang, L. Investigating bisulfite short-read mapping failure with hairpin bisulfite sequencing data. *BMC Genom.* **16**, S2 (2015).
48. Wang, H. et al. Sexual epigenetics: genome-wide analysis revealed differential DNA methylation in the vector tick *Haemaphysalis longicornis*. *Parasites Vectors*. **18**, 202 (2025).
49. Pagano, L. et al. Evaluation of Illumina and Oxford Nanopore Sequencing for the Study of DNA Methylation in Alzheimer's Disease and Frontotemporal Dementia. *Int. J. Mol. Sci.* **26**, 4198. https://doi.org/10.3390/ijms26094198 (2025).
50. Sigurpalsdottir, B. D. et al. A comparison of methods for detecting DNA methylation from long-read sequencing of human genomes. *Genome Biol.* **25**, 69. https://doi.org/10.1186/s13059-024-03207-9 (2024).
51. McArthur, C., Sanson, G. D. & Beal, A. M. Salivary proline-rich proteins in mammals: Roles in oral homeostasis and counteracting dietary tannin. *J. Chem. Ecol.* **21**, 663–691 (1995).
52. Bennick, A. Salivary proline-rich proteins. *Mol. Cell. Biochem.* **45**, 83–99 (1982).
53. Sugiyama, M. et al. Transcriptomic and functional screening of weapon formation genes implies significance of cell adhesion molecules and female-biased genes in broad-horned flour beetle. *PLoS Genet.* **19**, e1011069. https://doi.org/10.1371/journal.pgen.1011069 (2023).
54. Héberlé, É. & Bardet, A. F. Sensitivity of transcription factors to DNA methylation. *Essays Biochem.* **63**, 727–741 (2019).
55. Shayevitch, R., Askayo, D., Keydar, I. & Ast, G. The importance of DNA methylation of exons on alternative splicing. *RNA* **24**, 1351–1362 (2018).
56. Loughland, I., Little, A. & Seebacher, F. DNA methyltransferase 3a mediates developmental thermal plasticity. *BMC Biol.* **19**, 11 (2021).
57. Teets, N. M. & Denlinger, D. L. Physiological mechanisms of seasonal and rapid cold-hardening in insects. *Physiol. Entomol.* **38**, 105–116 (2013).
58. Zhan, X. et al. Arabidopsis proline-rich protein important for development and abiotic stress tolerance is involved in microRNA biogenesis. *Proc. Natl. Acad. Sci.* **109**, 18198–18203 (2012).
59. Engelhardt, J., Scheer, O., Stadler, P. F. & Prohaska, S. J. Evolution of DNA methylation across ecdysozoa. *J. Mol. Evol.* **90**, 56–72 (2022).
60. Agwunobi, D. O. et al. DNA Methyltransferases Contribute to Cold Tolerance in Ticks *Dermacentor silvarum* and *Haemaphysalis longicornis* (Acari: Ixodidae). *Front. Vet. Sci.* 8; (2021). https://doi.org/10.3389/fvets.2021.726731
61. Kotsarenko, K. et al. Newly identified DNA methyltransferases of *Ixodes ricinus* ticks. *Ticks Tick. Borne Dis.* **11**, 101348. https://doi.org/10.1016/j.ttbdis.2019.101348 (2020).
62. Kotsarenko, K. et al. Newly identified DNA methyltransferases of *Ixodes ricinus* ticks. *Ticks Tick. Borne Dis.* **11**, 101348 (2020).
63. Omar, M. A. A. et al. The Roles of DNA Methyltransferases 1 (DNMT1) in Regulating Sexual Dimorphism in the Cotton Mealybug, *Phenacoccus solenopsis*. *Insects* **11**, 121. https://doi.org/10.3390/insects11020121 (2020).
64. Deng, H., Jin, G. & Lim, B. Unveiling dynamic enhancer-promoter interactions in *Drosophila melanogaster*. *Biochem. Soc. Trans.* **50**, 1633–1642 (2022).
65. Moore, L. D., Le, T. & Fan, G. D. N. A. Methylation and Its Basic Function. *Neuropsychopharmacology* **38**, 23–38 (2013).
66. Ronai, I. et al. The repetitive genome of the *Ixodes ricinus* tick reveals transposable elements have driven genome evolution in ticks. *Genome Biol* (2026).
67. Petersen, M. et al. Diversity and evolution of the transposable element repertoire in arthropods with particular reference to insects. *BMC Ecol. Evol.* **19**, 11 (2019).
68. Artemov, A. V. et al. Genome-Wide DNA Methylation Profiling Reveals Epigenetic Adaptation of Stickleback to Marine and Freshwater Conditions. *Mol. Biol. Evol.* **34**, 2203–2213 (2017).
69. Tufts, D. M. et al. Phenotypic plasticity in blood-oxygen transport in highland and lowland deer mice. *J. Exp. Biol.* **216**, 1167–1173 (2013).
70. Flores, K. B., Wolschin, F. & Amdam, G. V. The role of methylation of DNA in environmental adaptation. *Integr. Comp. Biol.* **53**, 359–372 (2013).
71. Venney, C. J., Anastasiadi, D., Wellenreuther, M. & Bernatchez, L. The Evolutionary Complexities of DNA Methylation in Animals: From Plasticity to Genetic Evolution. *Genome Biol. Evol.* 15; (2023). https://doi.org/10.1093/gbe/evad216
72. McCaw, B. A., Stevenson, T. J. & Lancaster, L. T. Epigenetic Responses to Temperature and Climate. *Integr. Comp. Biol.* **60**, 1469–1480 (2020).
73. Gowri, V. & Monteiro, A. Inheritance of acquired traits in insects and other animals and the epigenetic mechanisms that break the Weismann barrier. *J. Dev. Biology*. **9**, 41 (2021).
74. Hu, J. et al. The epigenetic signature of colonizing new environments in Anolis lizards. *Mol. Biol. Evol.* **36**, 2165–2170 (2019).
75. Brevik, K., Bueno, E. M., McKay, S., Schoville, S. D. & Chen, Y. H. Insecticide exposure affects intergenerational patterns of DNA methylation in the Colorado potato beetle, *Leptinotarsa decemlineata*. *Evol. Appl.* **14**, 746–757 (2021).
76. Lyko, F. & Maleszka, R. Insects as innovative models for functional studies of DNA methylation. *Trends Genet.* **27**, 127–131 (2011).
77. de la Villalba, M., Piskobulu, V., Murgatroyd, C. & Hager, R. DNA methylation patterns respond to thermal stress in the viviparous cockroach *Diploptera punctata*. *Epigenetics* **16**, 313–326 (2021).
78. Kucharski, R., Maleszka, J., Foret, S. & Maleszka, R. Nutritional control of reproductive status in honeybees via DNA methylation. *Science* **319**, 1827–1830 (2008).
79. Alhosin, M. Epigenetics Mechanisms of Honeybees: Secrets of Royal Jelly. *Epigenet Insights*. **16**, 25168657231213717. https://doi.org/10.1177/25168657231213717 (2023).

80. Cassens, J. et al. Whole Genome Sequencing Reveals Clade-Specific Genetic Variation in Blacklegged Ticks. *Ecol. Evol.* **15**, e70987. https://doi.org/10.1002/ece3.70987 (2025).
81. Lane, R., Piesman, J. & Burgdorfer, W. Lyme borreliosis: relation of its causative agent to its vectors and hosts in North America and Europe. (1991).
82. Ginsberg, H. S. et al. Selective host attachment by *Ixodes scapularis* (Acari: Ixodidae): tick–lizard associations in the southeastern United States. *J. Med. Entomol.* **59**, 267–272 (2022).
83. 83 Jia, N. et al. Large-scale comparative analyses of tick genomes elucidate their genetic diversity and vector capacities. *Cell*. 182, 1328–1340. e1313 (2020).
84. Diuk-Wasser, M. A. et al. Spatiotemporal patterns of host-seeking *Ixodes scapularis* nymphs (Acari: Ixodidae) in the United States. *J. Med. Entomol.* **43**, 166–176 (2006).
85. Diuk-Wasser, M. A. et al. Field and climate-based model for predicting the density of host-seeking nymphal *Ixodes scapularis*, an important vector of tick-borne disease agents in the eastern United States. *Global Ecol. Biogeogr.* **19**, 504–514 (2010).
86. MacIntosh, G. H., Nuyens, A. C., Vickery, J. L., Berthold, A. & Lloyd, V. K. Epigenetic responses in Borrelia-infected Ixodes scapularis ticks: Over-expression of euchromatic histone lysine methyltransferase 2 and no change in DNA methylation. *PLoS One*. **20**, e0324546. https://doi.org/10.1371/journal.pone.0324546 (2025).
87. Cabezas-Cruz, A. et al. Anaplasma phagocytophilum increases the levels of histone modifying enzymes to inhibit cell apoptosis and facilitate pathogen infection in the tick vector *Ixodes scapularis*. *Epigenetics* **11**, 303–319 (2016).
88. Adamson, S. W., Browning, R. E., Budachetri, K., Ribeiro, J. M. & Karim, S. Knockdown of selenocysteine-specific elongation factor in *Amblyomma maculatum* alters the pathogen burden of *Rickettsia parkeri* with epigenetic control by the Sin3 histone deacetylase corepressor complex. *PLoS One*. **8**, e82012. https://doi.org/10.1371/journal.pone.0082012 (2013).
89. Krueger, F. et al. FelixKrueger/TrimGalore: v0.6.10 - add default decompression path (0.6.10). *Zenodo* https://doi.org/10.5281/zenodo.7598955 (2023).
90. Krueger, F. & Andrews, S. R. Bismark: a flexible aligner and methylation caller for Bisulfite-Seq applications. *Bioinformatics* **27**, 1571–1572 (2011).
91. Akalin, A. et al. methylKit: a comprehensive R package for the analysis of genome-wide DNA methylation profiles. *Genome Biol.* **13**, R87. https://doi.org/10.1186/gb-2012-13-10-r87 (2012).
92. Posit team. RStudio: Integrated Development Environment for R. Posit Software, PBC, Boston, MA. (2026). http://www.posit.co/
93. Gu, Z., Gu, L., Eils, R., Schlesner, M. & Brors, B. circlize Implements and enhances circular visualization in R. *Bioinformatics* **30**, 2811–2812 (2014).
94. Bushnell, B. BBduk v39.11. (2025). https://sourceforge.net/projects/bbmap/
95. Li, H. & Durbin, R. Fast and accurate short read alignment with Burrows–Wheeler transform. *Bioinformatics* **25**, 1754–1760 (2009).
96. Li, H. et al. The Sequence Alignment/Map format and SAMtools. *Bioinformatics* **25**, 2078–2079 (2009).
97. Institute, B. Picard toolkit v3.2.0 (2019). https://broadinstitute.github.io/picard/
98. Garrison, E. & Gabor, M. *Haplotype-based variant detection from short-read sequencing.* Preprint at (2012). https://arxiv.org/abs/1207.3907
99. O'Leary, S. J., Puritz, J. B., Willis, S. C., Hollenbeck, C. M. & Portnoy, D. S. These aren't the loci you'e looking for: Principles of effective SNP filtering for molecular ecologists. *Mol. Ecol.* **27**, 1–14. https://doi.org/10.1111/mec.14792 (2018).
100. Ord, J., Gossmann, T. I. & Adrian-Kalchhauser, I. High Nucleotide Diversity Accompanies Differential DNA Methylation in Naturally Diverging Populations. *Mol. Biol. Evol.* **40**, msad068. https://doi.org/10.1093/molbev/msad068 (2023).
101. Danecek, P. et al. The variant call format and VCFtools. *Bioinformatics* **27**, 2156–2158 (2011).
102. Lawrence, M. et al. Software for Computing and Annotating Genomic Ranges. *PLoS Comput. Biol.* **9**, e1003118. https://doi.org/10.1371/journal.pcbi.1003118 (2013).
103. Vogt, G. *Epigenetics in crustaceans.* in Epigenetics in aquaculture. (ed *Han-Ping Wang Francesc Piferrer)* 355–381 (2023). (Wiley Online Library.
104. Li, E. & Zhang, Y. DNA methylation in mammals. *Cold Spring Harb Perspect. Biol.* **6**, a019133. https://doi.org/10.1101/cshperspect.a019133 (2014).
105. Tuckow, A. P. & Temeyer, K. B. Discovery, adaptation and transcriptional activity of two tick promoters: Construction of a dual luciferase reporter system for optimization of RNA interference in *Rhipicephalus* (*Boophilus*) *microplus* cell lines. *Insect Mol. Biol.* **24**, 454–466 (2015).
106. Hernandez, E. P., Kusakisako, K., Hatta, T. & Tanaka, T. Characterization of an iron-inducible *Haemaphysalis longicornis* tick-derived promoter in an *Ixodes scapularis*-derived tick cell line (ISE6). *Parasit. Vectors.* **12**, 321. https://doi.org/10.1186/s13071-019-3574-9 (2019).
107. Kusakisako, K. et al. Transcriptional activities of two newly identified *Haemaphysalis longicornis* tick-derived promoter regions in the Ixodes scapularis tick cell line (ISE6). *Insect Mol. Biol.* **27**, 590–602 (2018).
108. Mi, H. et al. Protocol Update for large-scale genome and gene function analysis with the PANTHER classification system (v.14.0). *Nat. Protoc.* **14**, 703–721 (2019).
109. Mi, H. & Thomas, P. PANTHER pathway: an ontology-based pathway database coupled with data analysis tools. *Methods Mol. Biol.* **563**, 123–140 (2009).
110. Wickham, H. *ggplot2: Elegant Graphics for Data Analysis* (Springer-, 2016).
111. Keirans, J. E. & Clifford, C. M. The Genus Ixodes in the United States: A Scanning Electron Microscope Study and Key to the Adults1. *J. Med. Entomol.* **15**, 1–38 (1978).
112. Cooley, R. A. & Kohls, G. M. *The genus Ixodes in North America* (eds Robert Allen Cooley & Glen M. Kohls) 241–243Govt. Print. Off., (1945).
113. Steinig, E. C. Lachlan. Nanoq: ultra-fast quality control for nanopore reads. *J. Open. Source Softw.* **7**, 2991. https://doi.org/10.21105/joss.02991 (2022).
114. Flack, N. et al. The genome of Przewalski's horse (Equus ferus przewalskii). *G3: Genes Genomes Genet.* **14** (jkae113). https://doi.org/10.1093/g3journal/jkae113 (2024).
115. Li, H. Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics* **34**, 3094–3100 (2018).
116. Oxford Nanopore Technologies Plc. *Modkit v* 0.4.1 (2023). https://github.com/nanoporetech/modkit
117. Wreczycka, K. et al. Strategies for analyzing bisulfite sequencing data. *J. Biotechnol.* **261**, 105–115 (2017).
118. Quinlan, A. R. & Hall, I. M. BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics* **26**, 841–842 (2010).
119. Gel, B. & Serra, E. karyoploteR: an R/Bioconductor package to plot customizable genomes displaying arbitrary data. *Bioinformatics* **33**, 3088–3090 (2017).
120. Lyko, F. The DNA methyltransferase family: a versatile toolkit for epigenetic regulation. *Nat. Rev. Genet.* **19**, 81–92 (2018).
121. Swiss Institute of Bioinformatics. Expasy Translate tool (2025). https://web.expasy.org/translate/
122. Integrated DNA Technology Inc. OligoAnalyzerTM Tool in IDT (2025). https://www.idtdna.com/pages/tools/
123. Holman, L., Trontti, K. & Helanterä, H. Queen pheromones modulate DNA methyltransferase activity in bee and ant workers. *Biol. Lett.* **12**, 20151038. https://doi.org/10.1098/rsbl.2015.1038 (2016).
124. Biergans, S. D., Galizia, G., Reinhard, C., Claudianos, C. & J. & Dnmts and Tet target memory-associated genes after appetitive olfactory training in honey bees. *Sci. Rep.* **5**, 16223. https://doi.org/10.1038/srep16223 (2015).

125. Schmittgen, T. D. & Livak, K. J. Analyzing real-time PCR data by the comparative CT method. *Nat. Protoc.* **3**, 1101–1108 (2008).
126. Oliva Chávez, A. S. et al. Tick extracellular vesicles enable arthropod feeding and promote distinct outcomes of bacterial infection. *Nat. Commun.* **12**, 3696. https://doi.org/10.1038/s41467-021-23900-8 (2021).

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Author contributions

Conceptualization: AOC and JDO; Methodology: AOC, JDO, PS, MCG, and CF; Investigation: SGV, JC, PS, AL, EL, CH, BLG, CC, CW, SO, MT, TJ, NAM, and AOC; Formal analysis: SGV, JC, PS, CF, and AOC; Validation: SGV and JC; Resources: PS, TJ, RFM, DMT, JDO, and AOC; Visualization: SGV and JC; Writing – original draft preparation: SGV and JC; Writing – review and editing: PS, JC, BLG, MT, TJ, MCG, RFM, DMT, CF, JDO and AOC; Supervision: JDO, CF, and AOC; Project administration: JDO and AOC; Funding acquisition: JDO and AOC.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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